

Integration of YOLOv8 Object Detection in Crime Scene Investigation

Abstract— Object detection applies across diverse fields, including Crime Scene Investigation. Photographing evidence at crime scenes is a crucial process. Object detection and scene analysis can integrate into this procedure. However, the investigation process encompasses various stages. This study uses object detection for significant evidence found at crime scenes and related objects commonly observed in assault cases of NIJ reports. The study adopts transfer learning techniques, leveraging the pre-trained to train exceptional evidence at crime scenes. Consequently, this technique may contribute to developing a project plan for a Crime Scene Investigation model and more evidence object detection in the future. The tools utilized in this research include PyTorch, YOLOv8, and the LabelImg annotator. The acquisition method acquires them from online police news reports with 494 images. The custom dataset will train and fine-tune, leading to experimental results showcasing a high Mean Average Precision(mAP) in all target's custom datasets. These results indicate the potential use of object detection using YOLOv8 in crime scene investigation. Finally, this experiment shows how it detects imitation guns and related objects.

Keywords— YOLOv8, Object Detection, Crime Scene Investigation, Forensic Science

I. INTRODUCTION

This paper uses an artificial intelligence system for object detection to identify evidence in crime scene investigations(CSI). When a crime occurs, the investigating officer must thoroughly examine the crime scene. The photographic procedure should take photographs before the evidence gathering. The photographic documentation captures the environment in detail and establishes the connection between the evidence. This process cloud records footage before and after the crime scene inspection. Notably, the evidence encountered at crime scenes often differs from ordinary objects. For example, a knife may bear multiple traces of blood. Finally, the investigator must report the document to the court.

The benefits of implementing object detection are preventing the officer from danger, facilitating the investigator, avoiding evidence contamination, and analyzing the related evidence. For example, the system must advise the officer and seek the shell casing when it finds a gun on the scene. These objects may contain evidence or pose other hazards. In some instances, it may discover that the perpetrator has intentionally placed dangerous things near the crime scene to mislead investigators or harm officers. However, the examination will only cover some construction of the scene map in this experiment. Therefore, it is necessary to employ deep learning techniques to identify objects as forensic evidence. Those might transfer learning cover to the CSI model in the future. In this experiment, object detection

would detect extraordinary objects, such as imitation guns and related objects.

II. RELATED WORK

Object detection has demonstrated its versatility by using applications across diverse fields, such as crime scene investigation in forensic science [1]. It utilizes the Faster R-CNN algorithm in this context. It is significantly enhancing object detection accuracy on multiple datasets [2]. Moreover, a comprehensive survey conducts to gain valuable insights into various object detection techniques and their wide-ranging applications [3],[4]. These surveys explore the challenges, opportunities, and future research directions, including the intriguing area of crime prediction [5]. Crime scene investigation involves identifying general objects and specific items that may appear unusual compared to everyday objects, such as biological evidence, tool marks, and traces. Many systems have focused on using deep learning CNN-based approaches for detecting and analyzing evidence in crime scene images related to crime [6]. However, it depends on the technique and limitations of tools and methods specific to the evidence type. For example, surveillance systems use YOLOv3 to detect and classify guns [7]. Some work evaluates datasets containing images of firearms[8],[9][10]. However, the experiment aims to advance object detection techniques in forensic science, specifically in crime scene investigation. This section lead to forensic science basic and how object detection can be applied to it.

A. Evidence and Crime Scene Investigation

Evidence refers to information, facts, or data that supports or contradicts claims, assumptions, or beliefs. Evidence is pieces through experiments, observations, analysis, and rigorous research methods in scientific research. It serves to validate theories and confirm their validity. In a legal context, evidence supports or challenges specific assumptions and may be presented in court to prove or disprove a defendant's guilt [11]. In forensic science, evidence encompasses diverse types of physical [11], biological [12], trace [13], and digital evidence[14], subjected to multiple analysis techniques to support investigations. These objects may include weapons used in the crime, bloodstains, and changes observed in both the perpetrator and the victim, such as the condition of clothing and items related to the criminal case. Fortunately, The National Institute of Justice (NIJ) provides statistical data reports on crime evidence [15]. So, we use this report to reference assault cases at the crime scene and related types of weapons object detection (aka physical evidence).

B. Traditional Object Detection

Traditional object detection techniques use simple feature extraction and machine learning algorithms. Popular methods

include 1) The Viola-Jones algorithm, Which uses the properties of haarcascade, and the AdaBoost classifier to detect objects as a fast and efficient algorithm but less accurate than modern methods[16]. 2) HOG (Histogram of Oriented Gradients), It extracts features based on the slope values of the image. It is fast and efficient, but there are problems with complex textures and shapes objects [17]. 3) SIFT (Scale-Invariant Feature Transform), It extracts variety of characteristics based on changes in scale, rotation, and brightness is more effective for image variations. However, it requires high computational power [18]. As traditional methods widely use, advancements in deep learning have introduced new techniques for object detection. For example, face detection can perform using haarcascades, but deep learning-based classification methods can be employed instead of the traditional method.

C. Artificial Intelligence and Object Detection

Deep learning-based object detection is widespread due to its ability to handle large datasets with extensive annotations and the availability of powerful GPUs for training. Among the various deep learning techniques such as 1) Region-based Convolutional Neural Networks (R-CNN) is a hybrid of region and convolutional neural network in object detection [19]. 2) You Only Look Once (YOLO) is known for its speed and efficiency. it is reasonable for real-time applications requiring quick responses [20]. 3) Single Shot Detector (SSD) uses a series of convolutional layers to capture features at multiple scales, enabling the detection of objects of varying sizes[21].

However, YOLOv8 is an easy machine-learning API to learn and utilize. It suits imitation guns and related physical evidence in crime scene investigations. It has several advantages over other machine learning libraries. It is flexible, faster and provides optimizations that make it easier to train models. YOLOv8 can be an API or library with programming code and a command line [10].

D. Object Detection Dataset

The object detection dataset is the training and evaluation source of machine learning. This dataset typically consists of annotated images that contain objects of interest, along with annotations and information about the location and class of each object. 1) COCO (Common Objects in Context) is contain a huge of images dataset, diverse objects and detailed annotations, providing context and information about various object classes [22]. 2) PASCAL VOC (PASCAL Visual Object Classes) is a standardized dataset in object detection and classification[23]. It is used for evaluation and comparison of different methods. 3) ImageNet is a vast dataset that includes images covering a wide range of object categories [24]. 4) Open Images is a large-scale dataset created by Google, containing millions of images with annotations for object detection and classification. [25]. 5) The KITTI is dataset for autonomous driving. It provides annotations available for detecting objects such as cars, pedestrians, and cyclists [26],[27],[28]. 6) Caltech Pedestrian Detection is specifically designed for person detection[29],[30],[31].

This experiment uses the COCO format for custom dataset. However, we focus on evidence commonly found at crime scenes, considering statistics on assault cases provided by the National Institute of Justice (NIJ)[15]. The experiment includes various types of evidence that are easily detectable

by object detection systems, such as a variety of imitation guns, which require preprocessing and custom object selection to develop the crime scene investigation model.

E. Object Detection Models

Object detection models can be saved and deployed using various frameworks, depending on the specific machine learning framework used for training and deployment. Such as 1) TensorFlow is an open-source framework commonly used for machine learning and deep learning tasks. The model is known as Protocol Buffers [32]. 2) Keras is a machine learning API that runs on top of TensorFlow. The model is called Hierarchical Data Format (HDF5) [33]. 3) PyTorch is an open-source library for machine learning frameworks. The model is a Python pickle. It is a serialized and deserialized Python object structure [34]. It is a popular one. 4) ONNX (Open Neural Network Exchange) is a model of open-source format for representing models of machine learning. The model is a binary format as onnx [35]. 5) Caffe is a deep learning framework developed by Berkeley AI Research. The model also has a binary format as prototext or caffemodel [36]. 6) Scikit-learn is a popular machine learning library for Python known as the pickle file model. It has many APIs and libraries for machine learning[37].

This research utilize PyTorch and YOLOv8 that introducing a unified approach using a single neural network to process the images. The algorithm divides grid of cells on images and applies convolutional neural networks (CNNs) to each cell [10]. The objective is to develop a comprehensive object detection system using artificial intelligence to analyze preliminary case information directly from the crime scene. It suits the condition of physical evidence. Because YOLOv8 may not perform well with tiny-sized objects, and the results may be less accurate for such cases [38],[39].

III. METHODOLOGY

To initial the project, the authors categorized different types of evidence based on assault case report from the United States Department of Justice[15]. We select categories of physical evidence and specify imitation guns and related objects. Then used a pre-trained model “yolov8n.pt to perform transfer learning. The model supplement with additional dataset annotated using the Labelling tool in the COCO format. However, the research specifically selects high-frequency and prominent evidence types for object detection, as outlined in Table 1.

TABLE I. EVIDENCE CATEGORIES OF NIJ ASSAULT CASE

Evidence type	Example
Biological	Bloodstain etc.
Latent Prints	Fingerprint on object surface etc.
Pattern Evidence	Tools/marks etc.
Guns/Weapons	Guns/Weapons ^a etc.
Natural/Synthetic Materials	Clothing etc.

^a. Imitation guns and related objects is instance of physical evidence type Guns/Weapons.

Imitation firearms are frequently used in fatal crimes worldwide, making it challenging to investigate their sources. By utilizing images or video annotation. To enable machine learning techniques it mark and specify text for training and object detection. A crucial aspect of the experiment involves acquiring images from police news report websites and crime

scene evidence. The learning system's scope and the creation of a learning model are determined in Fig. 1.

Since preparing dataset encompassing all real-world objects for deep learning is impractical, the focus is placed on objects and evidence frequently encountered in crime scenes that may be limited. However, the experiment follows the framework, and the preparation steps in a computer vision technique for object detection are in Fig.1.

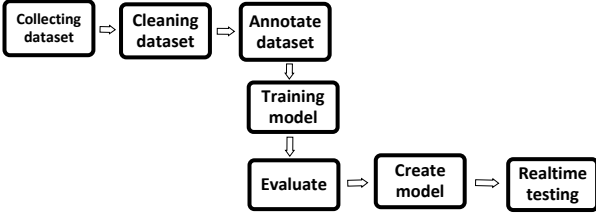


Fig. 1. Object detection framework

A. Collecting dataset

The steps of experiment start with collecting dataset of imitation gun and related from police's report news on website see Fig.2. This experiment creates a python function to download but they have other objects. So, the next step need to cleaning data.



Fig. 2. Imitation guns and related objects custom dataset from website

B. Cleaning dataset

Fig. 1 shows imitation guns and unusual weapon dataset used for the training model by selecting the related images. Imitation guns and related objects are composed of 7 classes.

['PenGun', 'ImitationShotGun', 'ShotGunBullet', 'Bullets', 'Bullet', 'ImitationRevolverGun', 'ImitationRifle']

The experiment retrieved imitation guns and related images 494 samples 32.4 MB for annotated objects. One image may have more than one object. LabelImg identifies the object location on the image and saves information in a text file. See a figure in the annotation step.

C. Annotating dataset

The annotation step is an important step to identify target objects. The imitation guns and related objects are defined in COCO format. We use "LabelImg"[40] Open-Source tools to annotate samples of images. We create directories of train dataset, test dataset and validate dataset. Put all annotated images to each directory. Finally, create "data.yaml" point to source of dataset.



Fig. 3. Custom dataset annotated example

Fig. 3 is examples of imitation guns with COCO format. After annotating the image, we will find labels in text file of annotated images and type of classes identification. We follow ultralytics documentation using command lines on custom dataset.

D. Training process

Training process based on our custom dataset. We use the "yolov8n.pt" weight model and input the training images and corresponding annotations into the model. To adjust the model's weights through an optimization process and save the best model for later utilization. It depends on parameters such as task, mode, model. it easy to use and utilized for this experiment. We need to use the "supported task" of YOLOv8 for inference, validation, and training. It depends on our tasks, we use "yolov8n.pt" to support this experiment and fine-tune for the CSI model in next step.

Meanwhile, the training process verifies that the value of the loss functions decreases and the average precision increases. Finally, it becomes stable. The box loss is obtaining the best intersection over union (IOU) between the object bounding box and prediction. Object loss is a function that helps predict the bounding box and determine the best fit during inference. According to Fig. 4, the bounding box of mAP50 and mAP50-95 of all class trends are positive.

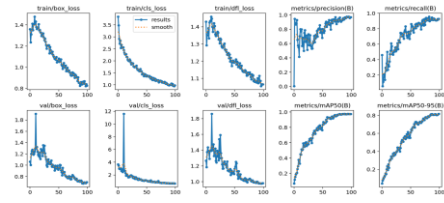


Fig. 4. Training chart after 100 epochs

E. Evaluate model

After initializing the "yolov8n.pt" is a supplement model as the pre-trained weights dataset. In this step, the model learns to adjust weights to the specific features of our dataset and the objects in the target that we need to detect. It will update to the "best.pt" model. However, YOLO evaluates metrics of object detection models based on mean average precision (mAP). This serves as a reliable measure to compare different models, evaluate improvements, and fine-tune hyperparameters in the YOLO algorithm.

- **Intersection Over Union (IOU)** employ for determining the correct prediction of the bounding box. It indicates how much bounding boxes overlap [41].

$$IOU = \frac{area(B_p \cap B_{gt})}{area(B_p \cup B_{gt})} \quad (1)$$

where:

- B_{gt} is ground truth bounding box
- B_p a predicted bounding box

- **Mean Average Precision(mAP)** is a performance and evaluation object detection based on Confusion Matrix, Recall, Precision, and Intersection over Union (IOU). YOLOv8

also uses mAP to evaluate the accuracy and performance of their models.

In the COCO object detection evaluation document, AP and mAP are the same methods. However, mAP is calculated by finding Average Precision (AP) for each class and then averaging over classes as follows.[42]

$$mAP = \frac{1}{N} \sum_{i=0}^N AP_i \quad (2)$$

where:

- N is number of classes
- AP_i is the average precision of class i

For instance, we can see Fig.5 and 6 of training results focus on mAP50 and mAP50-95.

```
Ultralytics YOLOv8.0.130 Python-3.9.12 torch-2.0.1cpu CPU
Model summary (Fused): 248 layers, 1132279 parameters, 0 gradients
```

	Class	Images	Instances	Box(P)	R	mAP50	mAP50-95	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
	all	40	161	0.955	0.996	0.994	0.892	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	PenGun	40	28	0.985	1	0.995	0.921	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	InitiationShotGun	40	35	0.972	0.996	0.994	0.932	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	ShotGunBullet	40	32	0.964	0.973	0.991	0.843	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	Bullet	40	23	0.988	1	0.995	0.786	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	Bullet	40	26	0.946	1	0.994	0.873	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	InitiationRevolverGun	40	4	0.984	1	0.995	0.971	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	InitiationKnife	40	13	0.96	1	0.995	0.93	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862

Speed: 1.6ms preprocess, 264.8ms inference, 0.8ms loss, 0.9ms postprocess per image

Fig. 5. Transection of custom dataset training

```
Ultralytics YOLOv8.0.130 Python-3.9.16 torch-2.0.1 CUDA-0 (NVIDIA GeForce RTX 3060, 12288MB)
Model summary (Fused): 148 layers, 100713 parameters, 0 gradients
```

	Class	Images	Instances	Box(P)	R	mAP50	mAP50-95	100%	90%	80%	70%	60%	50%	40%	30%	20%	10%	0%
	all	40	161	0.97	0.996	0.974	0.817	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	PenGun	40	28	1	0.994	0.995	0.864	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	InitiationShotGun	40	35	1	0.984	0.995	0.9	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	ShotGunBullet	40	32	1	0.899	0.983	0.782	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	Bullet	40	23	0.855	0.856	0.913	0.652	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	Bullet	40	26	0.993	0.962	0.973	0.8	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	InitiationRevolverGun	40	4	0.939	1	0.995	0.945	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862
	InitiationKnife	40	13	1	0.881	0.965	0.795	0.995	0.991	0.984	0.972	0.958	0.942	0.926	0.910	0.894	0.878	0.862

Speed: 1.2ms preprocess, 9.1ms inference, 0.8ms loss, 5.1ms postprocess per image

Fig. 6. Evaluation of all custom dataset all classes at mAP50 and mAP50-95

The mAP50 and mAP50-95 are performance and evaluate model measurement metrics. The mAP50 is the mean average precision at threshold of 0.5 of IOU, while the mAP50-95 is thresholds ranging from 0.5 to 0.95 of IOU. A higher mAP50 or mAP50-95 indicates better performance of an object detection model[41].

F. Create CSI model

After fine-tuning the new model, It automatic save the last fine-tune model as “best.pt”. The application can create a user interface to utilize the best CSI model later. We can use an analogy with other evidence as the same method specific for crime scene investigation works.

IV. EXPERIMENT RESULTS

A. Training results

The training started with supported task “yolov8n.pt” (nano-size model). In the local experiment, GPU Nvidia RTX 3060 Series was used. The training time depends on hardware, size, epoch and volume of image. This work defines 100 epochs of training with loss function both of train and validation dataset. The results trend is down of loss YOLO training measuring metrics in box_loss, cls_loss and dfl_loss in Fig. 4. Mean average precision(mAP) metrics of YOLO present in mAP50 and mAP50-95 that overall trends show good results for each epoch. See Fig. 5 and 6 training results.

B. Realtime Testing

In real-time testing, some of result low accuracy value of detection because the dataset does not cover various views for training. They acquire from the internet that shows imitation and so on the lineup on the table of the news reports. However, this experiment shows good testing results with precision values 0-1 (0-100%) in Fig.7.

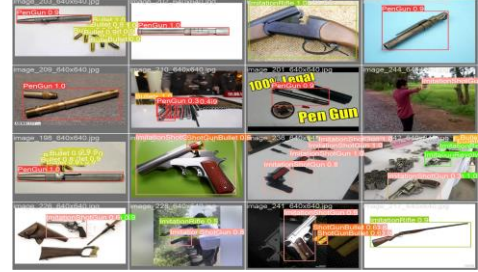


Fig. 7. Real-time testing

V. DISCUSSION AND CONCLUSION

YOLOv8 is a good choice to implement. It is robust object detection in real-time detection. However, it requires a significant amount of quality images and fine-tuning. Our custom dataset lined up on the table of the police news report with low qualities of images. YOLOv8 is best for images of 640x640 pixels, and the images may do densely packed or occluded because the image source varies when the size is adjusted for training.

The problem of the custom dataset is overlapping between imitation guns and other objects in the annotation process. So, the system train and detects as different objects, such as the background in Fig.5. It is the cause of the reduced real-time performance of this experiment. In Fig.8 shows a problem of annotation boundaries example.

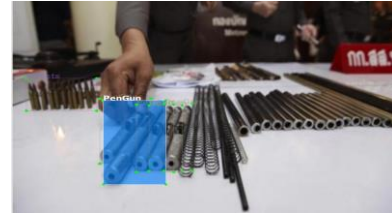


Fig. 8. The Overlapping of objects as the same class

However, the overall result shows in good trends and confidence enough to apply in crime scene investigation. It would be better if we do good annotation, high-quality, large, and diverse dataset. The performance of YOLOv8 are training fast and accurate. Therefore, it is appropriate for real-time applications, especially for crime scene investigation, and not tiny objects.

REFERENCES

- [1] S. Saikia, E. Fidalgo, E. Alegre, and L. Fernández-Robles, “Object Detection for Crime Scene Evidence Analysis Using Deep Learning,” in *Image Analysis and Processing - ICIAP 2017*, 2017, pp. 14–24.
- [2] S. Ren, K. He, R. B. Girshick, and J. Sun, “Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks,” *CoRR*, vol. abs/1506.01497, 2015.
- [3] L. Jiao, F. Zhang, F. Liu, S. Yang, L. Li, Z. Feng, and R. Qu, “A Survey of Deep Learning-Based Object Detection,” *IEEE Access*, vol. 7, pp. 128837–128868, 2019.
- [4] S. S. A. Zaidi, M. S. Ansari, A. Aslam, N. Kanwal, M. N. Asghar, and B. Lee, “A Survey of Modern Deep Learning based Object Detection Models,” *CoRR*, vol. abs/2104.11892, 2021.

- [5] V. Mandalapu, L. Elluri, P. Vyas, and N. Roy, "Crime Prediction Using Machine Learning and Deep Learning: A Systematic Review and Future Directions," *IEEE Access*, vol. 11, pp. 60153–60170, 2023.
- [6] M. Nakib, R. T. Khan, M. S. Hasan, and J. Uddin, "Crime Scene Prediction by Detecting Threatening Objects Using Convolutional Neural Network," in *2018 International Conference on Computer, Communication, Chemical, Material and Electronic Engineering (ICAME2)*, 2018, pp. 1–4.
- [7] S. Narejo, B. Pandey, D. Esenarro Vargas, C. Rodriguez, and M. Anjum, "Weapon Detection Using YOLO V3 for Smart Surveillance System," *Mathematical Problems in Engineering*, vol. 2021, pp. 1–9, 2021.
- [8] DaSCI, "Weapon Detection." University de Granada, Universidad de Jaen and Universidad de Cordoba, 2023.
- [9] V. V, C. R. K., and R. A. C., "Real Time Object Detection System with YOLO and CNN Models: A Review." 2022.
- [10] Ultralytics, "Ultralytics YOLOv8 Docs." Ultralytics, 2023.
- [11] NIJ, "Evidence Analysis and Processing." National Institute of Justice, 2023.
- [12] NIJ, "Biological Evidence." National Institute of Justice, 2023.
- [13] NIJ, "Overview of Trace Evidence." National Institute of Justice, 2023.
- [14] NIJ, "Digital Evidence and Forensics." National Institute of Justice, 2023.
- [15] J. Peterson, I. J. Sommers, D. Baskin, and D. Johnson, "The Role and Impact of Forensic Evidence in the Criminal Justice Process." National Institute of Justice, 2010.
- [16] P. Viola and M. Jones, "Rapid object detection using a boosted cascade of simple features," in *Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition. CVPR 2001*, 2001, vol. 1, pp. 1–I.
- [17] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR2005)*, 2005, vol. 1, pp. 886–893 vol. 1.
- [18] D. G. Lowe, "Object recognition from local scale-invariant features," in *Proceedings of the Seventh IEEE International Conference on Computer Vision*, 1999, vol. 2, pp. 1150–1157 vol.2.
- [19] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into High Quality Object Detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 6154–6162.
- [20] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 779–788.
- [21] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single shot multibox detector," in *European conference on computer vision*, 2016, pp. 21–37.
- [22] T.-Y. Lin, M. Maire, S. Belongie, L. Bourdev, R. Girshick, J. Hays, P. Perona, D. Ramanan, C. L. Zitnick, and P. Dollár, "Microsoft coco: Common objects in context," in *European conference on computer vision*, 2014, pp. 740–755.
- [23] M. Everingham, S. A. Eslami, L. Van Gool, C. K. Williams, J. Winn, and A. Zisserman, "The pascal visual object classes challenge: A retrospective," *International journal of computer vision*, vol. 111, no. 1, pp. 98–136, 2015.
- [24] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and F.-F. Li, "ImageNet: A Large-Scale Hierarchical Image Database." 2009.
- [25] Google, "Open Images Dataset V6." Google, 2023.
- [26] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, "Vision meets robotics: The KITTI dataset," *International Journal of Robotics Research (IJRR)*, vol. 32, no. 11, pp. 1231–1237, 2013.
- [27] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? the KITTI vision benchmark suite," in *Conference on Computer Vision and Pattern Recognition*, 2012, pp. 3354–3361.
- [28] M. Menze and A. Geiger, "Object Scene Flow for Autonomous Vehicles," in *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015.
- [29] P. Dollar, C. Wojek, B. Schiele, and P. Perona, "Pedestrian Detection: An Evaluation of the State of the Art," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 34, no. 4, pp. 743–761, 2012.
- [30] P. Dollar, C. Wojek, B. Schiele, and P. Perona, "Pedestrian detection: A benchmark," in *2009 IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 304–311.
- [31] W. Sun, S. Zhu, X. Ju, and D. Wang, "Deep learning based pedestrian detection," 2018, pp. 1007–1011.
- [32] C. Currier, "Protocol Buffers," in *Mobile Forensics – The File Format Handbook: Common File Formats and File Systems Used in Mobile Devices*, C. Hummert and D. Pawlaszczyk, Eds. Cham: Springer International Publishing, 2022, pp. 223–260.
- [33] Q. Koziol, "HDF5," in *Encyclopedia of Parallel Computing*, D. Padua, Ed. Boston, MA: Springer US, 2011, pp. 827–833.
- [34] N. S. Ketkar, "Introduction to PyTorch," in *Deep Learning with Python*, 2021.
- [35] J. Bai, F. Lu, K. Zhang, and others, "ONNX: Open Neural Network Exchange," *GitHub repository*. GitHub, 2019.
- [36] Y. Jia, E. Shelhamer, J. Donahue, S. Karayev, J. Long, R. B. Girshick, S. Guadarrama, and T. Darrell, "Caffe: Convolutional Architecture for Fast Feature Embedding," *arXiv preprint arXiv:1408.5093*, 2014.
- [37] C. Albon, *Machine Learning with Python Cookbook: Practical Solutions from Preprocessing to Deep Learning*. O'Reilly Media, Inc., 2018.
- [38] D. R. Don, R. Aygun, and M. Karakaya, "A Multistage Framework for Detection of Very Small Objects," in *The 6th International Conference on Machine Vision and Application (ICMVA 2023)*, 2023, pp. 9–14.
- [39] A. I. V. Amoguis, H. A. Hermida, G. J. B. Madrid, G. C. Marquez, J. G. O. Guerrero, and J. P. Ilao, "Road Lane Segmentation Using Vehicle Trajectory Tracking and Lane Demarcation Lines," in *The 6th International Conference on Machine Vision and Application (ICMVA 2023)*, 2023, pp. 64–71.
- [40] H. Labs, "LabelImg." GitHub, 2023.
- [41] R. Padilla, W. L. Passos, T. L. B. Dias, S. L. Netto, and E. A. B. da Silva, "A Comparative Analysis of Object Detection Metrics with a Companion Open-Source Toolkit," *Electronics*, vol. 10, no. 3, p. 279, 2021.
- [42] A. Ahmed, X. Parra, and J. Ortega-Garcia, "A new approach for automated forensic trace evidence detection in crime scenes," *Forensic Science International*, vol. 321, p. 110711, 2021.